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FULL PAPER

Rhinogobius biwaensis, a new gobiid fish of the "yoshinobori" species complex, *Rhinogobius* spp., endemic to Lake Biwa, Japan

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Abstract The endemic lentic goby of Lake Biwa in Japan, which has hitherto been designated informally as *Rhinogobius* sp. BW, is formally described as *Rhinogobius biwaensis* sp. nov. It is distinguished from other members of the genus by the following combination of characters: vertebral count 26 (rarely 27); pectoral fin rays 18–22 (mostly 20–21); longitudinal scales 31–36 (mostly 31–34, mode 32); predorsal scales absent or few; D1 low with non-filamentous spines, all of almost same length except for shorter 1st and 6th (last) spines in both sexes; pelvic fin disk composed of thin membranes, its frenum undeveloped with a low, thin lamella; cheek unmarked; back and sides of body with 5–6 unclear dark saddles and 6–9 dark blotches; D1 lacking blotches but melanized to greater or lesser degree along rays; dorsal edge of pectoral base with one small, dark spot tinged with blue metallic luster; in breeding males, yellow on suboperculum and branchiostegal region and on basal part of anal fin, blue on medial part of under jaw.

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Introduction

Rhinogobius Gill 1859 fishes are freshwater gobies that are distributed widely in East and Southeast Asia. They show a high degree of regional differentiation and comprise more than 86 species (Table 1), including the new species described herein (Wu and Zhong 2008; Akihito et al. 2013; Suzuki et al. 2015; Eschmeyer 2017). Apart from *Rhinogobius similis* Gill 1859, the Japanese species of this genus are very closely related and form a species complex that used to be lumped under the Japanese name "yoshinobori". *Rhinogobius similis,* which was proved by Suzuki et al. (2015) to be a senior synonym of *Rhinogobius giurinus* (Rutter 1897), is the only species of *Rhinogobius* from Japan that does not belong to this "yoshinobori" species complex. The members of this "yoshinobori" species complex are referred to in this paper as "yoshinobori" species or "yoshinobori" fishes. At least 16 "yoshinobori" species are currently recognized in various parts of Japan, though nine of them have no scientific name as yet.

The genus *Rhinogobius* was erected by Gill (1859) when he described *Rhinogobius similis* from Japan. Afterwards, Lake Biwa was counted as one of its habitats as a result of confusion concerning this genus, and the name "yoshinobori" was cited as one of Japanese local names of *R. similis* (see Jordan and Snyder 1901a; Jordan and Snyder 1901b: 56–58; Jordan et al. 1913: 341–342). Tomiyama (1936: 68–70) lumped the more than ten nominal species that had been described by then and considered them to represent one polymorphic species, "yoshinobori". As a result of reexamination of the type specimen, *Gobius brunneus*

Table 1 Meristic counts for the species of *Rhinogobius* compared with *Rhinogobius biwaensis* sp. nov.

No.	Scientific names	Vertebral counts	Predorsal scales	Pectoral fin rays	Longitudinal scale counts	Other references
1	*R. biwaensis* sp. nov.[a]	26–27 (26)	0–1 (0)	18–22 (20–21)	31–36 (31-34)	This paper
2	*R. albimaculatus* Chen, Kottelat and Miller 1999a	28	6–11	15–16	29–31	
3	*R. boa* Chen and Kottelat 2005	27	11–12	16	30–33	
4	*R. changtinensis* Huang and Chen 2007	27	0–2	15–16	28–30	
5	*R. cheni* (Nichols 1931)	27	about 4	unknown	34–35	Chen et al. 2008; Koumans 1940
6	*R. chiengmaiensis* Fowler 1934	28	0	unknown	26–27	Chen et al. 2008
7	*R. davidi* (Sauvage and Dabry 1874)	28	0	14–16	30–31	Chen and Miller 1998
8	*R. duospilus* (Herre 1935b)	27	8–12	15–17	29–31	Chen et al. 2002
9	*R. filamentosus* (Wu 1939)	27	5–8	15–17	30–33	Chen et al. 2008; Luo 1989
10	*R. flumineus* (Mizuno 1960b)	27–28	5–7	15–18	34–35	Chen et al. 1999b
11	*R. genanematus* Zhong and Tzeng 1998	27	2–5	15	27–29	Chen and Fang 2006
12	*R. henryi* (Herre 1938)	28	0	unknown	30–32	Chen et al. 2008
13	*R. honghensis* Chen, Yang and Chen 1999c	28	0–4	16–17	32–34	
14	*R. lentiginis* (Wu and Zheng in Zheng and Wu 1985)	27	0	14–16	30–32	Chen and Miller 1998
15	*R. lindbergi* Berg 1933	27–28	0	19–20	30–32	Sakai et al. 2000
16	*R. lineatus* Chen, Kottelat and Miller 1999a	28	6–7	15–16	29–32	
17	*R. linshuiensis* Chen, Miller, Wu and Fang 2002	27–28	7–11	14–16	28–31	
18	*R. liui* Chen and Wu 2008	29	0	19	36–39	
19	*R. longyanensis* Chen, Cheng and Shao 2008	27	6–8	17–18	30–32	
20	*R. lungwoensis* Huang and Chen 2007	28	3–7	14–16	31–34	
21	*R. maculicervix* Chen and Kottelat 2000	28	0–3	15–17	29–32	
22	*R. mekongianus* (Pellegrin and Fang 1940)	28	4–6	15–17	29–31	Chen et al. 1999a
23	*R. milleri* Chen and Kottelat 2003	27	8–10	15–16	28–31	
24	*R. multimaculatus* (Wu and Zheng in Zheng and Wu 1985)	29	0	14–18	33–37	Chen et al. 1999b
25	*R. nammaensis* Chen and Kottelat 2003	28	5–7	15–18	30–32	
26	*R. niger* Huang Chen and Shao 2017	27	0	16–18	35–37	
27	*R. parvus* (Luo 1989)	27	0–4	15–16	28–30	Chen et al. 2008
28	*R. ponkouensis* Huang and Chen 2007	28	4–5	16–17	32–34	
29	*R. rubromaculatus* Lee and Chang 1996	27-28	9–13	15–17	29–32	Chen et al. 1999b
30	*R. sulcatus* Chen and Kottelat 2005	27-28	8–11	15–17	26–29	
31	*R. szechuanensis* (Tchang 1939)	27	0	16–17	30–34	Chen et al. 2008; Wu and Zhong 2008
32	*R. taenigena* Chen, Kottelat and Miller 1999a	27	10–11	15–16	29	
33	*R. vermiculatus* Chen and Kottelat 2003	28	7–11	15–18	32–34	
34	*R. wangchuangensis* Chen, Miller, Wu and Fang 2002	27	3–6	15–16	28–30	
35	*R. wangi* Chen and Fang 2006	27	8–9	16	26–27	
36	*R. wuyanlingensis* Yang, Wu and Chen 2008	27	7–9	17–18	30–32	
37	*R. xianshuiensis* Chen, Wu and Shao 1999b	27	3–7	15–16	29–32	
38	*R. yaoshanensis* (Luo 1989)	28	10–12	16–17	29–31	Chen et al. 2008

Table 1 continued

No.	Scientific names	Vertebral counts	Predorsal scales	Pectoral fin rays	Longitudinal scale counts	Other references
39	*R. brunneus* (Temminck and Schlegel 1845)	26	11–13	18–20	32–35	Oijen et al. 2011
40	*R. candidianus* (Regan 1908a)	25-26	9–19	17–19	34–38	Chen and Shao 1996
41	*R. delicatus* Chen and Shao 1996	26	10–14	17–19	32–36	
42	*R. fluviatilis* Tanaka 1925	26	13–18	20–22	32–38	
43	*R. formosanus* Oshima 1919	26	9–14	19–21	32–34	Chen and Shao 1996
44	*R. gigas* Aonuma and Chen 1996	26	5–16	20–23	35–39	
45	*R. henchuenensis* Chen and Shao 1996	26	12–16	18–21	34–37	
46	*R. kurodai* (Tanaka 1908)	26	6–15	18–20	28–31	Suzuki and Chen 2011
47	*R. lanyuensis* Chen, Miller and Fang 1998	26	13–17	19–21	33–35	
48	*R. leavelli* (Herre 1935a)	26	6–16	18–20	30–34	Huang and Chen 2007
49	*R. maculafasciatus* Chen and Shao 1996	26	8–12	18–20	30–32	
50	*R. nagoyae* Jordan and Seale 1906	26	12–18	19–22	32–35	Akihito et al. 2013
51	*R. nandujiangensis* Chen, Miller, Wu and Fang 2002	25–26	8–11	17–18	27–29	
52	*R. nantaiensis* Aonuma and Chen 1996	26	13–17	18–19	33–36	
53	*R. ogasawaraensis* Suzuki, Chen and Senou 2012	26	7–16	18–20	31–35	
54	*R. reticulatus* Li, Zhong and Wu 2007	26–27	3–6	15–17	27–29	
55	*R. rubrolineatus* Chen and Miller 2008	26	3–5	16–18	28–30	
56	*R. sagittus* Chen and Miller 2008	26	3–4	16–17	29–31	
57	*R. sangenloensis* Chen and Miller 2014	26	9–12	16–17	25–27	
58	*R. similis* Gill 1859	26	8–12	18–19	29–31	Suzuki et al. 2015
59	*R. variolatus* Chen and Kottelat 2005	26–27	8–11	15–16	26–27	
60	*R. virgigena* Chen and Kottelat 2005	26	10–12	16	30–32	
61	*R. zhoui* Li and Zhong 2009	26	10–12	16–18	29–31	
62	*R. aporus* (Zhong and Wu 1998)	26	0	14	28–31	
63	*R. changjiangensis* Chen, Miller, Wu and Fang 2002	26	2–3	16–17	27–30	
64	*R. wuyiensis* Li and Zhong 2007	26	0–4	15–18	30–31	
65	*R. bucculentus* (Herre 1927a)	unknown	8–10	unknown	40–44	
66	*R. philippinus* (Herre 1927a)	unknown	12–14	unknown	36–40	
67	*R. carpenteri* Seale 1910	unknown	[b]	unknown	36–40	Herre, 1927a
68	*R. bedfordi* (Regan 1908b)	unknown	[c]	unknown	36–38	
69	*R. cliffordpopei* (Nichols 1925)	unknown	0–4[d]	18–20	27–30	Luo 1989; Chu and Chen 1990
70	*R. flavoventris* Herre 1927b	unknown	0	unknown	26	
71	*R. schultzei* (Herre 1927a)	unknown	15	unknown	29	
72	*R. vexillifer* (Fowler 1937)	unknown	10	16	27	
73	*R. imfasciocaudatus* Nguyen and Vo 2005	unknown	4	I, 17	30–31	
74	*R. longipinnis* Nguyen and Vo 2005	unknown	10–13	18	29–31	
75	*R. shennongensis* (Yang and Xie 1983)	unknown	5–6	I, 17–18	31–33	
76	*R. fukushimai* Mori 1934	unknown	2–6	18–20	30–31	Wu and Zhong 2008
77	*R. sowerbyi* Ginsburg 1917	unknown	6–8	unknown	35–36	
78	*Rhinogobius* sp. DL	26	8–10	19–22	36–39	Akihito et al. 2013
79	*Rhinogobius* sp. CO	26	16–20	19–22	31–37	Akihito et al. 2013
80	*Rhinogobius* sp. MO	26	10–13	20–22	28–31	Akihito et al. 2013
81	*Rhinogobius* sp. YB	26	0–14	18–20	35–37	Akihito et al. 2013
82	*Rhinogobius* sp. BB	26	5–13	16–19	33–35	Akihito et al. 2013

Table 1 continued

No.	Scientific names	Vertebral counts	Predorsal scales	Pectoral fin rays	Longitudinal scale counts	Other references
83	*Rhinogobius* sp. KZ	26	8–12	19–21	33–35	Akihito et al. 2013
84	*Rhinogobius* sp. BF	26	15	20–22	32–35	Suzuki et al. 2010
85	*Rhinogobius* sp. TO	26	3–15	19–21	31–34	Suzuki and Sakamoto 2005
86	*Rhinogobius* sp. OM	25–26	1–8	18–22	31–38	Takahashi and Okazaki 2002

[a] Parenthetical values represent most frequently

[b] Sometimes a few

[c] Nape covered with small scales

[d] 0 in males, 2–4 in females

Temminck and Schlegel (1845: 142–143) proved to belong to *Rhinogobius,* and *R. similis* came to be considered a junior synonym of the *Rhinogobius brunneus* (see Koumans 1935: 149; Boeseman 1947: 123; Takagi 1962; but see Suzuki et al. 2015). *Rhinogobius brunneus* (Temminck and Schlegel 1845) thus was then, for a while, treated as the valid scientific name of the polymorphic species "yoshinobori" in Japan.

Mizuno (1960a, b) then found a goby fish resembling the then *R. brunneus* but laying larger eggs, and he described it as a new species, *Tukugobius flumineus* Mizuno 1960, which became *Rhinogobius flumineus* (Mizuno 1960) because Herre (1933) had already declared that *Tukugobius* Herre is a junior synonym of *Rhinogobius* Gill. Thereafter some color-pattern morphs and egg-size morphs of *R. brunneus* were recognized one after another. Studies on ecology, ethology, and molecular genetics eventually revealed that most of those morphs represent distinct species (Mizuoka 1967, 1968, 1974, 1978; Nishijima 1968; Nakayama 1975; Mizuno 1976, 1981; Uehara 1980; Iwata 1981; Mizuno et al. 1982; Masuda et al. 1989; Suzuki 1992, 1996; Katoh and Nishida 1994). Those Tomiyama (1936: 68–70) had lumped together were subdivided again. An effort to assign formal scientific names is ongoing (Oijen et al. 2011; Suzuki and Chen 2011), though morphological similarity among taxa, lost or ambiguous type material, and insufficient taxonomic descriptions have made it difficult to complete the task. The undescribed species were provisionally named with two-capital-letter alphabetic designations (Mizuno et al. 1989).

Akihito et al. (2000) identified 11 "yoshinobori" species in Japan: *R. flumineus*, *Rhinogobius* sp. DA, *Rhinogobius* sp. CB, *Rhinogobius* sp. LD, *Rhinogobius* sp. BI, *Rhinogobius* sp. DL, *Rhinogobius* sp. CO, *Rhinogobius* sp. MO, *Rhinogobius* sp. YB, *Rhinogobius* sp. BB, *Rhinogobius* sp. OR. Among them, *Rhinogobius* sp. OR was considered to be a species having much variation, including four morphs: "Toshoku", "Gi-toshoku", "Shinjiko", and "Shimahire". After that, by morphological comparison with the type specimen, one of the "yoshinobori" species complex called *Rhinogobius* sp. DA was shown to actually be *R. brunneus* (Temminck and Schlegel 1845) (see Oijen et al. 2011). Detailed reinvestigation of morphology by Suzuki and Chen (2011) also suggested that *Rhinogobius* sp. CB, the "Gi-toshoku" morph of *Rhinogobius* sp. OR, and *Rhinogobius* sp. LD are actually the nominal species *Rhinogobius nagoyae* Jordan and Seale 1906, *Rhinogobius kurodai* (Tanaka 1908), and *Rhinogobius fluviatilis* Tanaka 1925, respectively. Suzuki et al. (2012) described *Rhinogobius* sp. BI formally as *Rhinogobius ogasawaraensis* Suzuki, Chen and Senou 2012.

Akihito et al. (2013) then split *Rhinogobius* sp. OR sensu Akihito et al. (2000) into six species on the basis of preceding works. The lake–river migrating population of Lake Biwa, representative of "Toshoku" morph, was considered a distinct species, *Rhinogobius* sp. OM. The morph "Gi-toshoku" included two species: *R. kurodai*, which lives in lentic water bodies in Tokyo and Kanagawa Prefectures (Suzuki and Chen 2011), and *Rhinogobius* sp. KZ from the Kazusa hilly area of Chiba Prefecture (Uehara 1980). The "Shimahire" morph mainly distributed in stagnant freshwaters around the Seto Inland Sea was elevated to a distinct species, *Rhinogobius* sp. BF (Ohara et al. 2009; Suzuki et al. 2010). In addition, the two lentic populations from Lake Biwa (Takahashi 1990, 1997; Takahashi and Okazaki 2002; Takahashi and Ohara 2004) and from the Nobi–Okazaki Plain in the Nagoya area (Nishimura 2000; Suzuki and Sakamoto 2005), which had been treated as variations of the "Shimahire" morph, were recognized as two different species, respectively: *Rhinogobius* sp. BW (described herein as *Rhinogobius biwaensis* sp. nov.) and *Rhinogobius* sp. TO. As a result, the number of recognized, albeit often undescribed, "Yoshinobori" species in Japan has reached 16 (Akihito et al. 2013). This left the morph "Shinjiko", some populations included in the morph "Gi-toshoku", and some other unclassified

populations that had been included in *Rhinogobius* sp. OR sensu Akihito et al. (2000) unresolved taxonomically. On the other hand, in the recent studies of the genus, *R. kurodai* is compared with *Rhinogobius* sp. BF and rediagnosed by Suzuki et al. (in press).

Takahashi and Okazaki (2002) proposed to divide the "yoshinobori" species into four groups depending on their life history patterns, namely amphidromous, fluvial, lake–river migrating, and lentic fishes. Amphidromy is the original lifestyle of the "yoshinobori" species complex: larvae hatch in a river and drift down immediately to the sea, feed and grow there, come back into the river at the young-fish stage, then grow some more and reproduce in the river. The fluvial fish is non-diadromous and completes its life cycle by laying its comparatively large eggs in the upper reaches of rivers. The lake-river migrating lifestyle resembles amphidromous lifestyle: fish migrates between lake and its inflowing rivers, substituting the lake for the sea. The lentic fish lives in a lake or pond and does not undertake lake-river migrations. The life styles of latter three must cause genetic isolation more easily than amphidromous life style. In Lake Biwa, an ancient lake in Japan, there inhabit not only a lake–river migrating species, *Rhinogobius* sp. OM, but also lentic species, *Rhinogobius* sp. BW. In the present study, we belatedly provide a formal description of *Rhinogobius* sp. BW, the lentic "yoshinobori" species in Lake Biwa.

Materials and methods

The holotype, LBM 1210013432, was caught in a traditional "eri" fish trap (Hashimoto 1999, 2015:373) at a depth of approximately 15 m off Wanihama beach on the southwestern shore of Lake Biwa in what is now Wani-nakahama, Otsu City, on 16 July 1997, and fixed at the sites in 10% formalin (i.e. 3.5–3.8% formaldehyde). Two of 78 paratypes, OMNH-P 23927 and OMNH-P 23932, were caught by hand net at Ukawa, Takashima City on 25 June 2011, and photographed while they were fresh, and fixed in 10% formalin. The remaining 76 paratypes, which were caught in either dragnets or "eri" fish trap, were fixed at the sites in 10% formalin.

Measurements, counts, and observations of fixed specimens were made under a binocular microscope following Hubbs and Lagler (1964) and Akihito et al. (2013). Measurements (to the nearest 0.1 mm) were made on the entire type series. Standard length, snout length, and head length were measured from the tip of the upper lip. Bones, fin rays, and scales of 41 paratype specimens were examined after staining them with sodium alizarinsulfonate (Kawamura and Hosoya 1991). The holotype and 37 other paratypes were not stained to avoid damage to them. Bones of the holotype were examined using soft X-ray by "M-60" apparatus of SOFTEX Co., Ltd., and its fin rays and scales were examined without staining. Cephalic sensory canal system of the holotype and three paratypes were examined. The observation methods and the names used for parts of the cephalic sensory canal system followed Akihito et al. (2013). Some specimens of nameless unclassified populations of *Rhinogobius* that were compared morphologically with the new species are shown in Table 2. Their predorsal scales were counted without staining.

Abbreviations and locations of depositories of specimens are as follows: AMS (The Australian Museum, Sydney, New South Wales, Australia), FAKU (The Kyoto University Museum, Sakyo-ku, Kyoto, Japan), LBM (Lake Biwa Museum, Kusatsu, Shiga, Japan), NSMT (National Museum of Nature and Science, Tsukuba, Ibaraki, Japan), OMNH (Osaka Museum of Natural History, Higashi-Sumiyoshi-ku, Osaka, Japan). Scientific names, Japanese names, and provisional two-capital-letter alphabetic designations of "yoshinobori" species in Japan follow Akihito et al. (2000, 2013).

Rhinogobius biwaensis sp. nov.

(Japanese name: Biwa-yoshinobori) (Figs. 1, 2, 3, 4; Table 3)

Rhinogobius brunneus: Takahashi 1990: 225 (Lake Biwa, Shiga, Japan).

Biwa-yoshinobori (tentative name): Takahashi 1997: 128 (Lake Biwa, Shiga, Japan).

Rhinogobius sp. BW: Takahashi and Okazaki 2002: 333 (Hamabun, Maiko, Shimosakamoto, Misaki and Biwa-cho, Lake Biwa, Shiga, Japan); Tsujimoto et al. 2003: 17 (Omi-Maiko, Lake Biwa, Shiga, Japan); Takahashi and Ohara 2004: 100 (Ukawa shore and Minamihama, Lake Biwa, Shiga, Japan); Senou et al. 2004: 459 (Lake Biwa, Shiga, Japan); Suzuki et al. 2010: 11 (Ukawa, Lake Biwa, Shiga; a pond, Watarai, Mie; Yato R., Gotsu, Shimane; and Taniyama R., Imabari, Ehime; Japan); Mukai et al. 2012: 18 (Sugaura and Kitakomatsu, Lake Biwa, Shiga, Japan); Akihito et al. 2013: 1461 (secondary sources); Yamasaki et al. 2015: 23 (Imajyuku and Moriyama, Lake Biwa, Shiga, Japan); Mukai et al. 2015: 175 (Ogasatakahashi R., Kikugawa, Shizuoka; Asuri Dam Reservoir, Toyota, Aichi; Yokoyama Dam Reservoir, Ibigawa, Gifu; a pond, Watarai, Mie; Yubara Dam Reservoir, Maniwa, Okayama, Yato R., Gotsu, Shimane; and Taniyama R., Imabari, Ehime; Japan). Fujita and Uchiyama 2015: 447 (Lake Biwa, Shiga, Japan; and secondary sources).

Holotype. LBM 1210013432, adult male, 33.0 mm SL, 35.1N, 135.9E, Wanihama beach, Lake Biwa, Wani-Nakahama, Otsu City, Shiga Prefecture, Japan, 15 m depth,

Table 2 Morphological comparison between specimens of nameless unclassified populations of *Rhinogobius* spp. and *Rhinogobius biwaensis* sp. nov.

Locality	No. of specimens[1]	Approximate SL (mm)	Predorsal scales counts	Registration No.
Tate R(Yoshinogari, Saga Pref.)	7 + 25 half bodies	55	9–17	LBM 1210055745
Shibukuma R. (Miwa town, Yamaguchi Pref.)	27 + 17 half bodies	50	9–15	LBM 1210055754
Maruyama R. (Hyogo Pref.)	10 half bodies	60	11–14	LBM 1210055753
Pond in Heian Shrine[2] (Kyoto Pref.)	3	35	8–12	FAKU 203106–FAKU 203108
Marsh Akauragata (Nanao City, Ishikawa Pref.)	one half body	45	>7	LBM 1210055748
Wetland Sakata (Akatsuka, Nigata Pref.)	15	25	12–16	LBM 1210055934
Marsh Tazawanuma (Oishida town, Yamagata Pref.)	8	30	10–13	LBM 1210055747

[1] Half body means the left half cut in a midsagittal plane

[2] Supplied with water by canal from Lake Biwa

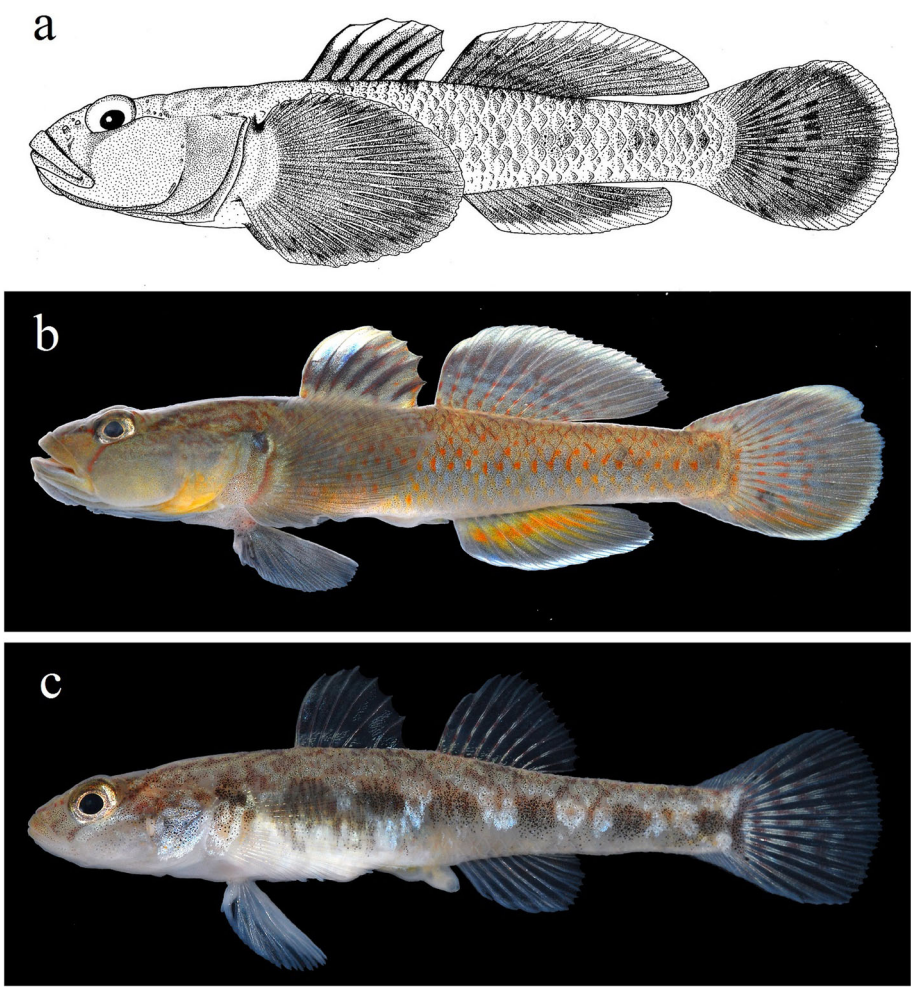

Fig. 1 Lateral views of *Rhinogobius biwaensis* sp. nov. **a** holotype, LBM 1210013432, male, 33.0 mm SL; **b** paratype, OMNH-P 23927, male, 36.0 mm SL; **c** paratype, OMNH-P 23932, female, 25.6 mm SL (b and c are photographed by T. Suzuki)

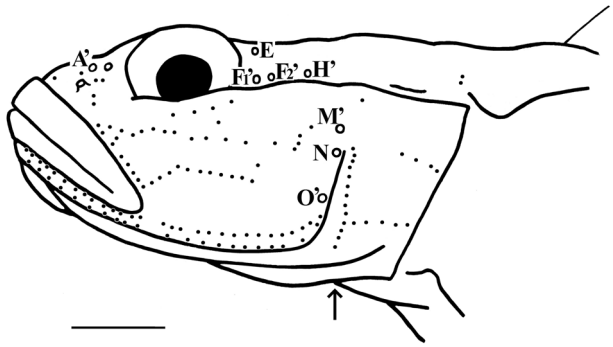

Fig. 2 Left lateral view of head of *Rhinogobius biwaensis* sp. nov., holotype, LBM 1210013432. Sensory canal pores (*A', E, F_1', F_2', H', M', N, O'*), sensory papillae (*small dots*), and the position where the gill membranes are attached to the isthmus (↑) is shown. *Scale bar* indicates 2 mm

"eri" fish trap, collected by S. Takahashi and S. Kobayashi, 16 July 1997.

Paratypes. Seventy-eight specimens. Two specimens of the 78, OMNH-P 23927 (a male) and OMNH-P 23932 (a female), were caught by T. Suzuki, M. Hosokawa and T. Yokoyama, with hand net in Lake Biwa at Ukawa, Takashima City on 25 June 2011. Seventy-six (40 males and 36 females) were caught in Lake Biwa (at Omi-Maiko in Minami-Komatsu, Otsu City; Shimosakamoto, Otsu City; and Misaki in Imahama-cho, Moriyama City) by S. Takahashi: NSMT-P55096 (3 males and 2 females), AMS I. 38802-001 (a male and a female), LBM 1210013433 (a female), LBM 1210028273–LBM 1210028307 (16 males and 19 females), Omi-Maiko, dragnet, 11 July 1990; LBM 1210013434 (a female), LBM 1210028310–LBM 1210028322 (10 males and 3 females), Misaki, "eri" fish trap, 23 July 1993; LBM 1210013435 (a male), LBM 1210028263 (a male), Omi-Maiko, dragnet, 10 June 1989; LBM 1210028257–LBM 1210028260 (2 males and 2 females), Omi-Maiko, dragnet, 25 July 1988; LBM 1210028261 (a male), Omi-Maiko, dragnet, 23 August 1988; LBM 1210028262 (a male), Omi-Maiko, dragnet, 03 September 1988; LBM 1210028264 (a male), Omi-Maiko, dragnet, 05 July 1989; LBM 1210028265–LBM 1210028272 (a male and 7 females), Shimosakamoto, "eri" fish trap, 02 May 1990; LBM 1210028308 (a male), LBM 1210028309 (a male), Omi-Maiko, dragnet, 22 May 1991.

Diagnosis. Vertebrae count 26 (rarely 27); pectoral fin rays 18–22 (mostly 20–21) longitudinal scales 31–36 (mostly 31–34, mode 32); predorsal scales absent or few (four of 40 paratypes had one predorsal scale); D1 low with non-filamentous spines, all of almost same length except for shorter 1st and 6th (last) spines in both sexes; pelvic fin disk composed of thin membranes, its frenum undeveloped with low, thin lamella; cheek unmarked, back and sides of

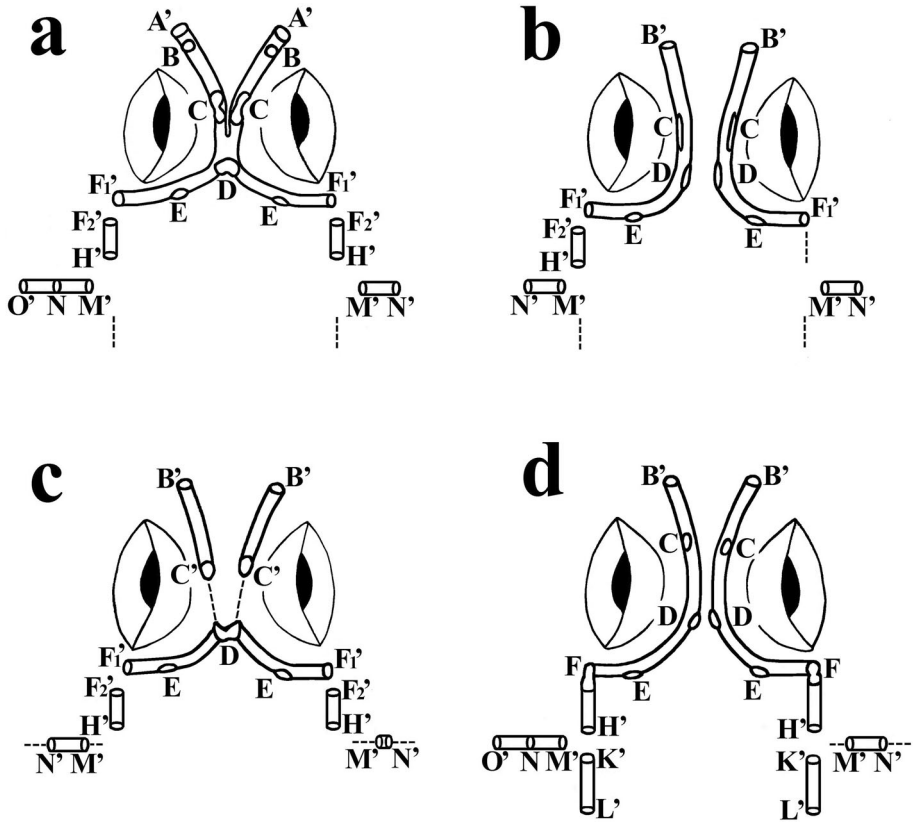

Fig. 3 Schematic illustration of cephalic sensory canal system of *Rhinogobius biwaensis* sp. nov. **a** holotype, LBM 1210013432, **b** paratype, LBM 1210013433, adult female, 23.3 mm SL, **c** paratype, LBM 1210013434, adult female, 27.0 mm SL, **d** paratype, LBM 1210013435, adult male, 32.9 mm SL. *The top* is anterior. *The capital letters* represent the sensory canal pores. *Pores A'–H'* anterior oculoscapular canal (*AOC*), *pores K'–L'* posterior oculoscapular canal (*POC*), pores *M'–O'* preopercular canal (*PC*). *The letters denoted by a prime mark such as A'* represent terminal pores. *The broken lines* indicate furrows

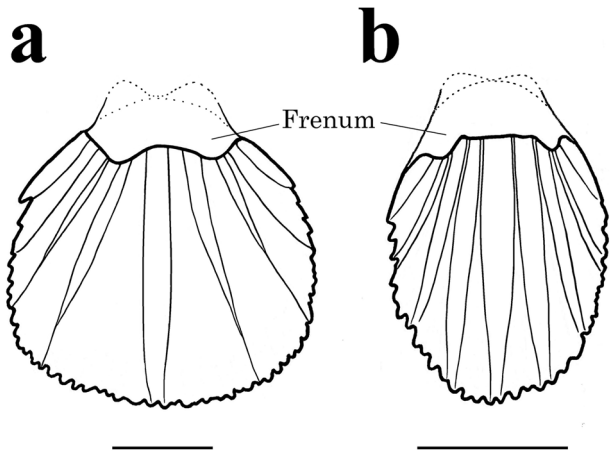

Fig. 4 Pelvic fins of *Rhinogobius biwaensis* sp. nov. **a** holotype, LBM 1210013432, **b** paratype, LBM 1210028284, adult male, 22.5 mm SL, *Scale bars* indicate 2 mm

body with 5–6 unclear dark saddles and 6–9 dark blotches; D1 lacking blotches but melanized to greater or lesser degree along rays; dorsal edge of pectoral base with one small, dark spot tinged with blue metallic luster; in breeding males, yellow on suboperculum and branchiostegal region and on basal part of anal fin, blue on medial part of under jaw.

Description. Measurements and counts for the type specimens are given in Table 3. The following description is based on the holotype, except as concerns sexual dimorphism and the cephalic sensory canal system. Variation seen in paratypes is given in parentheses.

Body. Elongate, subcylindrical anteriorly, compressed posteriorly. Vertebral count 10 + 16 = 26 (usually, but two cases of 10 + 17 = 27 and one case of 11 + 15 = 26 were also observed in 41 individuals), relationship between dorsal fin pterygiophores and vertebrae 3/22110/9 (usually, but 3/221100/10 in 1 of 36 individuals), number of anal pterygiophores anterior to first haemal spine 2 (2 or 3), number of epurals in holotype unknown owing to lack of clear radiograph (2 in paratypes).

Head. Large, somewhat depressed. Eyes dorsolateral. Snout blunt, very short. Mouth oblique, forming angle of about 35° with body axis. Posterior end of maxillary extending slightly beyond anterior margin of eye (reaching to vertical drawn through anterior margin of eye in most paratypes, but extending slightly beyond it in some males, falling slightly short of it in some females). On each jaw, 3–4 irregular rows of tiny, incurved, conical teeth dwindling to single row posteriorly. Larger teeth anteriorly and in outer row, though uneven. Tongue margin rounded, free anteriorly. Position of gill membrane attachment to isthmus shown in Fig. 2 (arrow).

Cephalic sensory canal system. On cheek, three longitudinal rows of sensory papillae, one row is directly below eye and others lower on middle of cheek: for details of holotypes see Fig. 2. Morphology of cephalic sensory canal system highly variable, as previously reported by Takahashi and Okazaki (2002), and as exemplified by present 4 specimens (cf. Fig. 3). In holotype (Figs. 2, 3a), anterior oculoscapular canals (hereinafter AOC) beginning at pore A', interrupted at pore F, this pore being divided into anterior (F_1') and posterior (F_2') pores that open on either side of gap in canal; pores C enlarged lengthwise; pore D single; posterior oculoscapular canals (POC) expressed as open furrows, not closed tubes; preopercular canals (PC) with 3 pores on left side (M', N, O'), 2 on right (M', N'). In LBM 1210013433 (Fig. 3b), AOC beginning at pore B', interrupted at pores F as in holotype; pores C and D opening as long slits, left and right pore D (and left and right AOC at that point) separated; posterior to pores F, AOC with 2 pores (F_2', H') on left side, just a furrow on right side; POC an indistinct furrow on both sides; PC with 2 pores on both sides. In LBM 1210013434 (Fig. 3c), AOC beginning at pore B', interrupted at pores F as in holotype, with parts between pores C and D expressed as open furrows and right and left pores D in contact; AOC present posterior to pores F (F_2'–H') on both sides; POC absent; PC with 2 pores on both sides, but with tubular part very short on right side. In LBM 1210013135 (Fig. 3d), AOC continuous from B' to H', pores D of right and left sides separated, pores F enlarged lengthwise; POC (K'–L') present on both sides; PC with 3 pores on left side, 2 on right.

Fins. Fin spines and rays as follows: D1 VI; D2 I, 8 (I, 7–9; mostly I, 8); anal fin I, 8 (I, 8–9; mostly I, 8); pectoral fin 21 (18–22; mostly 20–21); pelvic fin I, 5. Dorsal fins separate, D1 lower than D2. Spines of D1 non-filamentous and of almost same length except for shorter first and last spines. D1 reaching D2 when depressed (not reaching that far in small male paratypes and all female paratypes). D2 reaching caudal fin when depressed (not reaching that far in paratypes; larger males tending to have longer D2). Anal fin origin situated below second branched ray of D2. Anal fin lower than D2, not reaching caudal fin when depressed. Pectoral fin large, oblong (larger in males than in females). Pelvic fin disk-shaped, circular, composed of thin membranes with undeveloped frenum with low, thin lamella (Fig. 4a) (disk longitudinally more elliptical, e.g. Fig. 4b, in most paratypes, but rounded in some of largest specimens). The undeveloped frenum with low, thin lamella is a remarkable characteristic of this species, indicating that sucker disk is not developed and the fish does not adapt stream. Caudal fin rounded.

Scales. Trunk with ctenoid scales. Longitudinal scale count 32 (31–36; mostly 31–34, mode 32); transverse scale count 10 (8–11; mostly 9–10); predorsal scales absent (four of 40 paratypes had one predorsal scale). Area anterior to

Table 3 Proportional measurements and counts of *Rhinogobius biwaensis* sp. nov.

	Holotype LBM 1210013432	Paratypes	
	male	males	Females
		n = 41	*n* = 37
Standard length (mm)	33.0	21.0–43.1	20.2–33.0
Total length (mm)	41.0	26.1–52.6	24.8–39.9
Measurements as % of standard length			
		mean [range]	mean [range]
Body depth	19.7	20.3 [16.3–22.9]	20.9 [17.2–24.3]
Head length	32.1	31.1 [28.4–33.7]	29.9 [28.2–31.7]
Head depth	16.7	17.0 [15.4–19.4]	15.7 [14.1–17.3]
Head width	21.2	21.7 [17.6–26.3]	19.0 [17.2–20.7]
Snout length	9.1	9.2 [7.1–11.6]	8.2 [7.2– 9.3]
Postorbital head length	16.1	15.5 [14.3–16.7]	15.1 [13.7–16.6]
Orbit length	7.6	7.4 [6.1– 9.0]	7.7 [6.4– 8.9]
Upper jaw length	14.2	12.0 [9.5–14.5]	9.8 [9.0–10.9]
Gape width	15.2	14.6 [10.3–19.9]	11.3 [9.6–13.0]
Predorsal length	39.4	39.7 [37.1–42.2]	39.8 [38.2–43.2]
Preanal length	60.9	61.1 [59.2–62.9]	61.6 [59.7–65.3]
Caudal peduncle length	24.2	26.4 [25.0–29.3]	26.5 [25.1–28.5]
Caudal peduncle depth	10.9	10.5 [9.3–11.7]	10.1 [9.2–10.8]
1st dorsal fin base	17.6	16.3 [13.1–18.5]	16.1 [12.1–17.9]
2nd dorsal fin base	18.8	16.0 [13.7–18.0]	15.4 [13.8–17.3]
Anal fin base	15.5	13.6 [11.3–16.5]	13.2 [11.8–14.7]
Depressed 1st dorsal fin	21.5	19.7 [16.8–23.2]	18.0 [16.7–19.9]
Depressed 2nd dorsal fin	40.3	30.1 [23.2–38.0]	22.2 [20.3–24.1]
Depressed anal fin	30.6	25.4 [20.9–33.9]	20.5 [18.8–21.6]
Pectoral fin length	30.9	26.6 [23.3–30.5]	24.4 [19.6–26.4]
Pelvic fin length	18.8	19.3 [17.6–21.8]	19.3 [17.6–21.6]
Counts			
		value (*n*)	
Vertebrae	10 + 16 = 26	10 + 16 = 26 (38); 11 + 15 = 26 (1); 10 + 17 = 27 (2)	
Dorsal fin rays	VI-I,8	VI–I, 7 (2); VI–I, 8 (32); VI–I, 9 (7)	
Anal fin rays	I,8	I, 8 (39); I, 9 (2)	
Pelvic fin rays (P2)	I,5	I, 5 (41)	
P-V	3/22110/9	3/22110/9 (35); 3/221100/10 (1)	
AP	2	2 (16); 3 (8)	
EPU	not examined	2 (29)	
Pectoral fin rays (P1)	21	18 (1); 19 (2); 20 (18); 21 (17); 22 (3)	
Longitudinal scales	32	31 (8); 32 (13); 33 (7); 34 (5); 35 (3); 36 (1)	
Transverse scales	10	8 (3); 9 (20); 10 (12); 11 (3)	
Predorsal scales	0	0 (36); 1 (4)	

P-V relationship between dorsal fin pterygiophores and vertebrae; *AP* number of anal pterigiophores anterior to first hemal spine; *EPU* number of epurals

line connecting D1 origin, pectoral insertion, and anal origin scaleless. Rest of body covered with ctenoid scales, with a few cycloid scales on border with naked area.

Color in formalin. Ground color grayish brown; cheek unmarked; back and sides of body with five indistinct darker saddles and seven blotches (6–8 blotches in paratypes, clarity of this pattern and deepness of ground color diverse, but mature males usually darker); vertical fins melanized except for clear margin, areas along spines of D1 in particular being quite black (in all female and some male paratypes, vertical fins clearer, and D2 and caudal fin with 2–4 rows of small, dark spots on rays); one small, dark spot at dorsal edge of pectoral fin base, caudal fin base lacking dark spots (dark spot present in some female paratypes), anal fin lacking spots.

Color in life. In addition to color and melanization patterns noted above, the small dark spot at dorsal edge of pectoral fin base is tinged with blue metallic luster when alive. Iridescent colors from metallic yellow (top of fin) to metallic blue (toward the middle of fin) present on anterior margin of first dorsal fin, same iridescent sheen on edge of second dorsal fin and dorsal part of caudal fin, and bluish shiny coloration along edge of anal fin and ventral part of caudal fin. Anterior half of exposed part of each scale orangish-brown, imparting reticulated appearance to fish (all these colors much paler in females).

Other sexual dimorphism. Urogenital papilla distinctly pointed in males but rounded in females. D1 rounded in females and immature individuals, often rectangular in mature males. Mature males with yellow on suboperculum and branchiostegal regions and on basal part of anal fin, blue on medial part of under jaw.

Distribution. *Rhinogobius biwaensis* sp. nov. is endemic to Lake Biwa, Shiga Prefecture, Japan, being found in the lake proper mainly at depths of 10 to 30 m, rarely in the mouths or lowest reaches of inflowing rivers, and in lagoons connected to Lake Biwa; from May to August, they migrate to shallow water for spawning; all the habitats are effectively lentic. *Rhinogobius biwaensis* specimens have been found in some other regions of western Japan (Suzuki et al. 2010; Mukai et al. 2015). It is thought that they originated from individuals introduced artificially and accidentally from Lake Biwa, because their mtDNA haplotypes do not show geographical variation and are same or very close to the haplotypes of *R. biwaensis* from Lake Biwa (Mukai et al. 2015).

Remarks. *Comparison with nominal species of Rhinogobius*. We compared *R. biwaensis* with 76 valid or possibly valid nominal species (Nos. 2 to 77 in Table 1) of *Rhinogobius* from a wide area of East and Southeast Asia, in particular about vertebral counts (VC), predorsal scale counts (Pred. S), pectoral fin ray counts (P1), and longitudinal scale counts (LR), based on original descriptions and some reviewal works (Herre, 1927a; Koumans 1940; Luo 1989; Chen and Shao 1996; Chen and Miller, 1998, Chen et al. 1999a, 1999b; Sakai et al., 2000; Chen et al. 2002; Chen and Fang 2006; Huang and Chen 2007; Chen et al., 2008; Wu and Zhong 2008; Oijen et al., 2011; Suzuki and Chen, 2011; Akihito et al. 2013; Suzuki et al. 2015; Eschmeyer 2017). *Rhinogobius* species can be divided into two groups, those with low vertebral counts (VC: 25–27, mostly 26) and those with high vertebral counts (VC: 27–29). In Japan, only *Rhinogobius flumineus* has a high vertebral count (VC: 27–28), while its congeners, including *R. biwaensis*, belong to the group with low vertebral counts (Akihito et al. 2013). In contrast, outside Japan approximately half of the species of *Rhinogobius* have high vertebral counts. Chen et al. (2008) listed the 34 then-known nominal species of *Rhinogobius* with high vertebral counts and provided a diagnostic key to them. Including three species described later, the number of species of *Rhinogobius* with high vertebral counts is now 37 (Nos. 2 to 38 in Table 1), all of them differing from *R. biwaensis* in this respect.

Besides *R. biwaensis*, 26 valid nominal species of *Rhinogobius* are reported to have low vertebral counts. Twenty-three of them (Nos. 39–61 in Table 1) have three or more predorsal scales whereas one of the distinctive characteristics of *R. biwaensis* is its usually scaleless nape, although four specimens among examined 40 paratypes had one predorsal scale, and Takahashi and Okazaki (2002) reported the existence of one individual with three predorsal scales. Among the species of *Rhinogobius* with low vertebral counts, the remaining three species (Nos. 62–64), *Rhinogobius aporus* (Zhong and Wu 1998) (Pred. S: 0, P1: 14), *Rhinogobius changjiangensis* Chen, Miller, Wu and Fang 2002 (Pred. S: 2–3, P1: 16–17), and *Rhinogobius wuyiensis* Li and Zhong 2007 (Pred. S: 0–4, P1: 15–18), have no or few predorsal scales like *R. biwaensis*, but have lower pectoral fin ray counts than this new species does (P1: 18–22). As an additional distinctive difference, adult males of *R. wuyiensis* have filamentous extended rays of D1, while these rays are not extended in *R. biwaensis*.

The vertebral counts of the remaining 13 nominal species of *Rhinogobius* are unknown, but scale counts provide a means of differentiating *R. biwaensis* sp. nov. from them. Four of these species (Nos. 65–68 in Table 1) have 36 or more longitudinal scales. Four others of these 13 species (Nos. 69–72) have lower longitudinal scale counts of 30 or less. The remaining five species of the 13 (Nos. 73–77) have several to many predorsal scales. *Rhinogobius biwaensis*, with an LR of 31–36 (mostly 31–34, mode 32) and generally scaleless napes can, therefore, be distinguished easily from these 13 species, as well as from all its other known congeners as noted above.

Comparison with "yoshinobori" species in Japan. We made a morphological comparison of *R. biwaensis* with the other 15 "yoshinobori" species and some of the unclassified populations of *Rhinogobius* sp. OR sensu Akihito et al. (2000) in Japan. As Takahashi and Okazaki (2002) reported, *Rhinogobius* sp. OM in Lake Biwa is characterized by a round pelvic fin disk composed of thick membranes having a large thick frenum, existence of predorsal scales (1–8, mean 4.5), and triangular high D1 of adult male. These characteristics are shared by eleven other Japanese species: *R. flumineus, Rhinogobius brunneus*, *Rhinogobius fluviatilis*, *Rhinogobius nagoyae, Rhinogobius ogasawaraensis*, *Rhinogobius* sp. DL, *Rhinogobius* sp. CO, *Rhinogobius* sp. MO, *Rhinogobius* sp. YB, *Rhinogobius* sp. BB, and *Rhinogobius* sp. KZ (Akihito et al. 2013). *Rhinogobius biwaensis* differs from all of these because of having a pelvic fin disk composed of thin membranes with a undeveloped frenum (in many cases, the pelvic fin disk is longitudinally elliptical), a generally scaleless predorsal area, and a low D1 in both sexes. The remaining three species, *Rhinogobius kurodai* (Pred. S: 6–15), *Rhinogobius* sp. BF (Pred. S: 15), and *Rhinogobius* sp. TO (Pred. S: 3–15) resemble *R. biwaensis* in having a slender pelvic fin disk composed of thin membrane, an undeveloped frenum with a low lamella, and a low D1 in both sexes, but these three species have predorsal scales unlike *R. biwaensis* (Table 1). Such predorsal scales were also observed on specimens of the nameless unclassified populations of *Rhinogobius* sp. OR sensu Akihito et al. (2000) from both western and eastern Japan (Table 2). The fishes from Tate R., Shibukuma R., and Maruyama R. are "Shinjiko" morph sensu Akihito et al. (2000). Those from a pond of Heian Shrine and from marshes or wetland in central–eastern Japan are small, having indistinct morphological characteristics. Thus *R. biwaensis* is clearly distinguished not only from *R. kurodai, Rhinogobius* sp. BF, and *Rhinogobius* sp. TO, but also from all nameless unclassified populations which were examined, by the general absence of predorsal scales.

Based on allozyme data, Takahashi and Okazaki (2002) revealed that the two species, *R. biwaensis* and *Rhinogobius* sp. OM, in Lake Biwa water system are reproductively isolated. Takahashi and Ohara (2004) showed that mtDNA PCR-RFLP analysis can discriminate between *R. biwaensis* and *Rhinogobius* sp. OM even if specimens are small or in short supply. Allozyme analysis by the first author (unpublished) indicated that *R. biwaensis* is different from other populations of *Rhinogobius* sp. OR sensu Akihito et al. (2000) from various parts of Japan. Yamasaki et al. (2015) estimated the phylogenetic relationships among Japanese species of *Rhinogobius* by using multiple nuclear gene sequences. Their results showed that *R. biwaensis* is clearly distinguished from other species. *Rhinogobius biwaensis*, as well as most of the other "yoshinobori" species included in Akihito et al. (2013), was monophyletic in the nuclear gene trees. The unclassified populations that belong to *Rhinogobius* sp. OR sensu Akihito et al. (2000), however, did not appear to be monophyletic. *Rhinogobius biwaensis* and two other freshwater species, *Rhinogobius* sp. OM and *Rhinogobius* sp. BF, appeared to be closely related to each other and formed a clade in the trees, with no admixture from the unclassified populations. This result suggests that *R. biwaensis* is probably not an introduced species but it is likely to have evolved from a common ancestor with *Rhinogobius* sp. OM and *Rhinogobius* sp. BF.

Taking all these evidences together, *R. biwaensis* is clearly distinguished from other "yoshinobori" fishes of Japan both morphologically and phylogenetically, and its status as a distinct species deserving of a formal scientific name can no longer be doubted.

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